

LAYER: 3\_MesMDM

This file is a merged representation of a subset of the codebase, containing files not matching ignore patterns, combined into a single document by Repomix.

<file\_summary>

This section contains a summary of this file.

<purpose>

This file contains a packed representation of a subset of the repository's contents that is considered the most important context.

It is designed to be easily consumable by AI systems for analysis, code review, or other automated processes.

</purpose>

<file\_format>

The content is organized as follows:

1. This summary section
2. Repository information
3. Directory structure
4. Repository files (if enabled)
5. Multiple file entries, each consisting of:
  - File path as an attribute
  - Full contents of the file

</file\_format>

<usage\_guidelines>

- This file should be treated as read-only. Any changes should be made to the original repository files, not this packed version.
- When processing this file, use the file path to distinguish between different files in the repository.
- Be aware that this file may contain sensitive information. Handle it with the same level of security as you would the original repository.

</usage\_guidelines>

<notes>

- Some files may have been excluded based on .gitignore rules and Repomix's configuration
- Binary files are not included in this packed representation. Please refer to the Repository Structure section for a complete list of file paths, including binary files
- Files matching these patterns are excluded: \*\*/obj/\*\*, \*\*/bin/\*\*, \*\*/\*.png, \*\*/\*.jpg, \*\*/node\_modules/\*\*, \*\*/\*.Designer.cs
- Files matching patterns in .gitignore are excluded
- Files matching default ignore patterns are excluded
- Files are sorted by Git change count (files with more changes are at the bottom)

</notes>

</file\_summary>

<directory\_structure>

Promatis.Net.MES.MDM.Configuration/Class1.cs  
Promatis.Net.MES.MDM.Configuration/Promatis.Net.MES.MDM.Configuration.csproj  
Promatis.Net.MES.MDM.Data/DBContext/MesMdmApplicationDbContext.cs  
Promatis.Net.MES.MDM.Data/Promatis.Net.MES.MDM.Data.csproj  
Promatis.Net.MES.MDM.Data/Triggers/TechnologicalOperationHierarchyTrigger.cs  
Promatis.Net.MES.MDM.Domain.Interface/Class1.cs  
Promatis.Net.MES.MDM.Domain.Interface/Promatis.Net.MES.MDM.Domain.Interface.csproj  
Promatis.Net.MES.MDM.Domain/Promatis.Net.MES.MDM.Domain.csproj  
Promatis.Net.MES.MDM.Domain/TechnologicalOperation/TechnologicalOperation.cs  
Promatis.Net.MES.MDM.Domain/TechnologicalOperation/TechnologicalOperationValidator.cs  
Promatis.Net.MES.MDM.Domain/TechnologicalOperationParameter/TechnologicalOperationParameter.cs  
Promatis.Net.MES.MDM.Domain/TechnologicalOperationParameter/  
TechnologicalOperationParameterValidator.cs  
Promatis.Net.MES.MDM.Domain/TechnologicalOperationUnit/TechnologicalOperationUnit.cs  
Promatis.Net.MES.MDM.Domain/TechnologicalOperationUnit/TechnologicalOperationUnitValidator.cs  
Promatis.Net.MES.MDM.Domain/TechnologicalParameter/TechnologicalParameter.cs  
Promatis.Net.MES.MDM.Domain/TechnologicalParameter/TechnologicalParameterValidator.cs  
Promatis.Net.MES.MDM.Domain/TechnologicalParameterCalcMethod/TechnologicalParameterCalcMethod.cs  
Promatis.Net.MES.MDM.Domain/TechnologicalParameterCalcMethod/  
TechnologicalParameterCalcMethodValidator.cs  
Promatis.Net.MES.MDM.Domain/Units/DepartmentUnitBase.cs  
Promatis.Net.MES.MDM.Domain/Units/DepartmentUnitBaseValidator.cs  
Promatis.Net.MES.MDM.Domain/Units/PositionUnitBase.cs  
Promatis.Net.MES.MDM.Domain/Units/PositionUnitBaseValidator.cs  
Promatis.Net.MES.MDM.Domain/Units/ProductionUnitBase.cs  
Promatis.Net.MES.MDM.Domain/Units/ProductionUnitBaseValidator.cs  
Promatis.Net.MES.MDM.Domain/Units/StorageUnitBase.cs

```

Promatis.Net.MES.MDM.Domain/Units/StorageUnitBaseValidator.cs
Promatis.Net.MES.MDM.Domain/Units/TransportUnitBase.cs
Promatis.Net.MES.MDM.Domain/Units/TransportUnitBaseValidator.cs
Promatis.Net.MES.MDM.Service/Promatis.Net.MES.MDM.Service.csproj
Promatis.Net.MES.MDM.Service/Services/TechnologicalOperationParameterService.cs
Promatis.Net.MES.MDM.Service/Services/TechnologicalOperationService.cs
Promatis.Net.MES.MDM.Service/Services/TechnologicalParameterCalcMethodService.cs
Promatis.Net.MES.MDM.Service/Services/TechnologicalParameterService.cs
Promatis.Net.MES.MDM.Tests/Promatis.Net.MES.MDM.Tests.csproj
Promatis.Net.MES.MDM.Tests/UnitCategoryValidationTests.cs
Promatis.Net.MES.MDM.Tests/UnitTest1.cs
Promatis.Net.MES.MDM.UI/_Imports.razor
Promatis.Net.MES.MDM.UI/Pages/TechnologicalOperations/Card/TechnologicalOperationEditDialog.razor
Promatis.Net.MES.MDM.UI/Pages/TechnologicalOperations/Card/TechnologicalOperationEditDialog.razor.cs
Promatis.Net.MES.MDM.UI/Pages/TechnologicalOperations/TechnologicalOperationPage.razor
Promatis.Net.MES.MDM.UI/Pages/TechnologicalOperations/TechnologicalOperationPageContext.cs
Promatis.Net.MES.MDM.UI/Pages/TechnologicalParameter/Card/TechnologicalParameterDialog.razor
Promatis.Net.MES.MDM.UI/Pages/TechnologicalParameter/Card/TechnologicalParameterDialog.razor.cs
Promatis.Net.MES.MDM.UI/Pages/TechnologicalParameter/Card/TechnologicalParameterGeneralTab.razor
Promatis.Net.MES.MDM.UI/Pages/TechnologicalParameter/Card/TechnologicalParameterTechnicalTab.razor
Promatis.Net.MES.MDM.UI/Pages/TechnologicalParameter/Card/
TechnologicalParameterTechnicalTab.razor.cs
Promatis.Net.MES.MDM.UI/Pages/TechnologicalParameter/TechnologicalParameterPage.razor
Promatis.Net.MES.MDM.UI/Pages/TechnologicalParameter/TechnologicalParameterPageContext.cs
Promatis.Net.MES.MDM.UI/Promatis.Net.MES.MDM.UI.csproj
Promatis.Net.MES.MDM.UI/UiModule.cs
Promatis.Net.MES.MDM.UI/wwwroot/exampleJsInterop.js
</directory_structure>

```

<files>

This section contains the contents of the repository's files.

<file path="Promatis.Net.MES.MDM.Configuration/Class1.cs">

```
namespace Promatis.Net.MES.MDM.Configuration
```

```
{
```

```
    public class Class1
```

```
    {
```

```
    {
```

```
    }
}
```

```
}
```

</file>

<file path="Promatis.Net.MES.MDM.Configuration/Promatis.Net.MES.MDM.Configuration.csproj">

```
<Project Sdk="Microsoft.NET.Sdk">
```

```
{
```

```
  <PropertyGroup>
```

```
    <TargetFramework>net10.0</TargetFramework>
```

```
    <ImplicitUsings>enable</ImplicitUsings>
```

```
    <Nullable>enable</Nullable>
```

```
  </PropertyGroup>
```

```
{
```

```
  <ItemGroup>
```

```
    <ProjectReference Include="..\..\..\
```

```
\MesCore\Promatis.Net.MES.Configuration\Promatis.Net.MES.Configuration.csproj" />
```

```
    <ProjectReference Include="..\
```

```
\Promatis.Net.MES.MDM.Service\Promatis.Net.MES.MDM.Service.csproj" />
```

```
  </ItemGroup>
```

```
{
```

```
</Project>
```

</file>

<file path="Promatis.Net.MES.MDM.Data/DBContext/MesMdmApplicationDbContext.cs">

```
using Microsoft.EntityFrameworkCore;
```

```
using Microsoft.Extensions.Configuration;
```

```
using Promatis.Net.MES.Data;
```

```
using Promatis.Net.MES.MDM.Domain;
```

```
{
```

```
namespace Promatis.Net.MES.MDM.Data;
```

```
{
```

```
public class MesMdmApplicationDbContext(
```

```
    DbContextOptions options,
```

```
    IConfiguration configuration,
```

```
    IServiceProvider? serviceProvider = null) :
```

```
    MesApplicationDbContext(options, configuration, serviceProvider)
```

```
{
```

```
    public DbSet<TechnologicalOperation> TechnologicalOperations => Set<TechnologicalOperation>();
```

```

        public DbSet<TechnologicalOperationUnit> TechnologicalOperationUnits =>
Set<TechnologicalOperationUnit>();
        public DbSet<TechnologicalParameter> TechnologicalParameters => Set<TechnologicalParameter>();
        public DbSet<TechnologicalOperationParameter> TechnologicalOperationParameters =>
Set<TechnologicalOperationParameter>();
        public DbSet<TechnologicalParameterCalcMethod> TechnologicalParameterCalcMethods =>
Set<TechnologicalParameterCalcMethod>();
    }
</file>

<file path="Promatis.Net.MES.MDM.Data/Promatis.Net.MES.MDM.Data.csproj">
<Project Sdk="Microsoft.NET.Sdk">
    <PropertyGroup>
        <TargetFramework>net10.0</TargetFramework>
        <ImplicitUsings>enable</ImplicitUsings>
        <Nullable>enable</Nullable>
    </PropertyGroup>
    <ItemGroup>
        <PackageReference Include="Microsoft.EntityFrameworkCore" Version="10.0.8" />
        <PackageReference Include="Microsoft.EntityFrameworkCore.Relational" Version="10.0.8" />
        <PackageReference Include="Microsoft.Extensions.Configuration" Version="10.0.8" />
    </ItemGroup>
    <ItemGroup>
        <ProjectReference Include="..\..\..\MesCore\Promatis.Net.MES.Data\Promatis.Net.MES.Data.csproj" />
        <ProjectReference Include="..\Promatis.Net.MES.MDM.Domain\Promatis.Net.MES.MDM.Domain.csproj" />
    </ItemGroup>
</Project>
</file>

<file path="Promatis.Net.MES.MDM.Data/Triggers/TechnologicalOperationHierarchyTrigger.cs">
using Promatis.Net.MES.Data;
using Promatis.Net.MES.MDM.Domain;
namespace Promatis.Net.MES.MDM.Data;
/// <summary>
/// A C05Dt=Cä5 C0@C,,:C'0CD=Cä5 D 5D,,5C08CR 4C'0 D$@C,,3C45D 0 C,,5D 0D EC,,8 D$5DT=Cä;Cä3C,,GCTAC8DR >C05D 0Dd
/// </summary>
public class TechnologicalOperationHierarchyTrigger : TechnologicalOperationHierarchyTrigger<
    TechnologicalOperation,
    TechnologicalOperationUnit,
    TechnologicalOperationParameter>
{
}
</file>

<file path="Promatis.Net.MES.MDM.Domain.Interface/Class1.cs">
namespace Promatis.Net.MES.MDM.Domain.Interface
{
    public class Class1
    {
    }
}
</file>

<file path="Promatis.Net.MES.MDM.Domain.Interface/Promatis.Net.MES.MDM.Domain.Interface.csproj">
<Project Sdk="Microsoft.NET.Sdk">
    <PropertyGroup>
        <TargetFramework>net10.0</TargetFramework>
        <ImplicitUsings>enable</ImplicitUsings>
        <Nullable>enable</Nullable>
    </PropertyGroup>
    <ItemGroup>
        <ProjectReference Include="..\..\..\MesCore\Promatis.Net.MES.Domain.Interface\Promatis.Net.MES.Domain.Interface.csproj" />
    </ItemGroup>
</Project>

```

```
</file>
```

```
<file path="Promatis.Net.MES.MDM.Domain/Promatis.Net.MES.MDM.Domain.csproj">
<Project Sdk="Microsoft.NET.Sdk">
  <PropertyGroup>
    <TargetFramework>net10.0</TargetFramework>
    <ImplicitUsings>enable</ImplicitUsings>
    <Nullable>enable</Nullable>
  </PropertyGroup>
  <ItemGroup>
    <ProjectReference Include="..\..\..\MesCore\Promatis.Net.MES.Domain\Promatis.Net.MES.Domain.csproj" />
    <ProjectReference Include="..\Promatis.Net.MES.MDM.Domain.Interface\Promatis.Net.MES.MDM.Domain.Interface.csproj" />
  </ItemGroup>
</Project>
</file>
```

```
<file path="Promatis.Net.MES.MDM.Domain/TechnologicalOperation/TechnologicalOperation.cs">
using Promatis.Net.MES.Domain;
namespace Promatis.Net.MES.MDM.Domain;
public class TechnologicalOperation : TechnologicalOperationBase<
    TechnologicalOperation,
    TechnologicalOperationUnit,
    TechnologicalOperationParameter>
{
    // A;C AD 0C ACä;DäBCô> Dt8D B. D
    // Aä= C 2D$>CÄ0D$8Dt5D :C, ACä4CT@Cd8D" AC$>C"AD$2C &VçB 8 Children D BC,,?Cä< TechnologicalOperation
}
</file>
```

```
<file path="Promatis.Net.MES.MDM.Domain/TechnologicalOperation/TechnologicalOperationValidator.cs">
using Promatis.Net.MES.Domain;
namespace Promatis.Net.MES.MDM.Domain;
public class TechnologicalOperationValidator :
    TechnologicalOperationBaseValidator<TechnologicalOperation>
{
    D
}
</file>
```

```
<file path="Promatis.Net.MES.MDM.Domain/TechnologicalOperationParameter/
TechnologicalOperationParameter.cs">
using Promatis.Net.MES.Domain;
namespace Promatis.Net.MES.MDM.Domain;
public class TechnologicalOperationParameter :
    TechnologicalOperationParameterBase<TechnologicalOperation, TechnologicalParameter> { }
</file>
```

```
<file path="Promatis.Net.MES.MDM.Domain/TechnologicalOperationParameter/
TechnologicalOperationParameterValidator.cs">
using Promatis.Net.MES.Domain;
namespace Promatis.Net.MES.MDM.Domain;
public class TechnologicalOperationParameterValidator :
    TechnologicalOperationParameterBaseValidator<TechnologicalOperationParameter,
    TechnologicalOperation, TechnologicalParameter>
{
    D
}
</file>
```

```
<file path="Promatis.Net.MES.MDM.Domain/TechnologicalOperationUnit/TechnologicalOperationUnit.cs">
using Promatis.Net.MES.Domain;
namespace Promatis.Net.MES.MDM.Domain;
```

```

Ð
public class TechnologicalOperationUnit : TechnologicalOperationUnitBase<TechnologicalOperation>Ð
{Ð
    /// <summary>Ð
    /// Aô5D 5C¤@D´2C 5CÂ 1C 7Cä2Cä5 D 2Cä9D BC$> B #A    D BD >C4> D$8Cö8Ct8D >C$0C0=D´< D 2Cä9D BC$>CÂ 4C´0
    /// </summary>Ð
    public new virtual TechnologicalOperation OperationÐ
    {Ð
        get => base.Operation as TechnologicalOperation ?? null!;Ð
        set => base.Operation = value;Ð
    }Ð
}
</file>

<file path="Promatis.Net.MES.MDM.Domain/TechnologicalOperationUnit/
TechnologicalOperationUnitValidator.cs">
using Promatis.Net.MES.Domain;Ð
Ð
namespace Promatis.Net.MES.MDM.Domain;Ð
Ð
public class TechnologicalOperationUnitValidator :
TechnologicalOperationUnitBaseValidator<TechnologicalOperationUnit, TechnologicalOperation>Ð
{Ð
    Ð
}
</file>

<file path="Promatis.Net.MES.MDM.Domain/TechnologicalParameter/TechnologicalParameter.cs">
using Promatis.Net.MES.Domain;Ð
Ð
namespace Promatis.Net.MES.MDM.Domain;Ð
Ð
public class TechnologicalParameter : TechnologicalParameterBaseÐ
{Ð
    // Bt8D BD´9 C¤;C AD 4C´0 DbSet<TechnologicalParameter>Ð
}
</file>

<file path="Promatis.Net.MES.MDM.Domain/TechnologicalParameter/TechnologicalParameterValidator.cs">
using Promatis.Net.MES.Domain;Ð
Ð
namespace Promatis.Net.MES.MDM.Domain;Ð
Ð
public class TechnologicalParameterValidator :
TechnologicalParameterBaseValidator<TechnologicalParameter>Ð
{Ð
    Ð
}
</file>

<file path="Promatis.Net.MES.MDM.Domain/TechnologicalParameterCalcMethod/
TechnologicalParameterCalcMethod.cs">
using Promatis.Net.MES.Domain;Ð
Ð
namespace Promatis.Net.MES.MDM.Domain;Ð
Ð
/// <summary>Ð
/// A¤>C0:D 5D$=C 0 D 5C ;C,,7C FC,,0 CÄ5D$>CD0 D 0D GCTBC BCTEC0>C´>C48Dt5D :Cä3Câ ?C @C <CTBD 0.Ð
/// B 2Dô7D´2C 5D" :Cä=C¤@CTBC0CDâ >Cô5D 0Dd8DâÂ >C >D CCD>C$0C08CR 8 C00D 0CÄ5D$@ C,,7 C¤>C0BCT:D BC ÖW40D0i
/// </summary>Ð
public class TechnologicalParameterCalcMethod : TechnologicalParameterCalcMethodBase<UnitBase,
TechnologicalOperation, TechnologicalParameter>Ð
{Ð
    // A¤;C AD GC,,AD" 8 C4>D$>C" : CÄ0C0?C,,=C4C C" F$6öçFW#B <Cä4D4;Dò ÖW40D0i
    // B 2Cä9D BC$0 Unit, TechnologicalOperation C, FV6†æöÆöv-6 Å & ÖWFW"
    // C 2D$>CÄ0D$8Dt5D :C, ?Cä;D4GC,,;C, AD$@Cä3C,,5 C¤>C05Dt=D´5 D$8CöK.Ð
}
</file>

<file path="Promatis.Net.MES.MDM.Domain/TechnologicalParameterCalcMethod/
TechnologicalParameterCalcMethodValidator.cs">
using Promatis.Net.MES.Domain;Ð
Ð
namespace Promatis.Net.MES.MDM.Domain;Ð
Ð
public class TechnologicalParameterCalcMethodValidator :
TechnologicalParameterCalcMethodBaseValidator<Ð

```

```

        TechnologicalParameterCalcMethod, Ø
        UnitBase, Ø
        TechnologicalOperation, Ø
        TechnologicalParameter> Ø
    { Ø
    }
</file>

```

```

<file path="Promatis.Net.MES.MDM.Domain/Units/DepartmentUnitBase.cs">
using Promatis.Net.MES.Domain; Ø
using Promatis.Net.MES.Domain.Interface; Ø
Ø
namespace Promatis.Net.MES.MDM.Domain; Ø
Ø
public abstract class DepartmentUnitBase : UnitBase Ø
{ Ø
    protected DepartmentUnitBase() => Kind = UnitKind.Department; Ø
}
</file>

```

```

<file path="Promatis.Net.MES.MDM.Domain/Units/DepartmentUnitBaseValidator.cs">
using Promatis.Net.MES.Domain; Ø
Ø
namespace Promatis.Net.MES.MDM.Domain; Ø
Ø
public abstract class DepartmentUnitBaseValidator<T> : UnitBaseValidator<T> where T :
DepartmentUnitBase Ø
{ Ø
}
</file>

```

```

<file path="Promatis.Net.MES.MDM.Domain/Units/PositionUnitBase.cs">
using Promatis.Net.MES.Domain; Ø
using Promatis.Net.MES.Domain.Interface; Ø
Ø
namespace Promatis.Net.MES.MDM.Domain; Ø
Ø
public abstract class PositionUnitBase : UnitBase Ø
{ Ø
    protected PositionUnitBase() => Kind = UnitKind.Position; Ø
}
</file>

```

```

<file path="Promatis.Net.MES.MDM.Domain/Units/PositionUnitBaseValidator.cs">
using Promatis.Net.MES.Domain; Ø
Ø
namespace Promatis.Net.MES.MDM.Domain; Ø
Ø
public class PositionUnitBaseValidator<T> : UnitBaseValidator<T> where T : PositionUnitBase Ø
{ Ø
}
</file>

```

```

<file path="Promatis.Net.MES.MDM.Domain/Units/ProductionUnitBase.cs">
using Promatis.Net.MES.Domain; Ø
using Promatis.Net.MES.Domain.Interface; Ø
Ø
namespace Promatis.Net.MES.MDM.Domain; Ø
Ø
public abstract class ProductionUnitBase : UnitBase Ø
{ Ø
    protected ProductionUnitBase() => Kind = UnitKind.Production; Ø
}
</file>

```

```

<file path="Promatis.Net.MES.MDM.Domain/Units/ProductionUnitBaseValidator.cs">
using Promatis.Net.MES.Domain; Ø
Ø
namespace Promatis.Net.MES.MDM.Domain; Ø
Ø
public class ProductionUnitBaseValidator<T> : UnitBaseValidator<T> where T : ProductionUnitBase Ø
{ Ø
}
</file>

```

```

<file path="Promatis.Net.MES.MDM.Domain/Units/StorageUnitBase.cs">

```

```

using Promatis.Net.MES.Domain;
using Promatis.Net.MES.Domain.Interface;

namespace Promatis.Net.MES.MDM.Domain;

public abstract class StorageUnitBase : UnitBase
{
    protected StorageUnitBase() => Kind = UnitKind.Storage;
}
</file>

<file path="Promatis.Net.MES.MDM.Domain/Units/StorageUnitBaseValidator.cs">
using Promatis.Net.MES.Domain;

namespace Promatis.Net.MES.MDM.Domain;

public class StorageUnitBaseValidator<T> : UnitBaseValidator<T> where T : StorageUnitBase
{
}
</file>

<file path="Promatis.Net.MES.MDM.Domain/Units/TransportUnitBase.cs">
using Promatis.Net.MES.Domain;
using Promatis.Net.MES.Domain.Interface;

namespace Promatis.Net.MES.MDM.Domain;

public abstract class TransportUnitBase : UnitBase
{
    protected TransportUnitBase() => Kind = UnitKind.Transport;
}
</file>

<file path="Promatis.Net.MES.MDM.Domain/Units/TransportUnitBaseValidator.cs">
using Promatis.Net.MES.Domain;

namespace Promatis.Net.MES.MDM.Domain;

public class TransportUnitBaseValidator<T> : UnitBaseValidator<T> where T : TransportUnitBase
{
}
</file>

<file path="Promatis.Net.MES.MDM.Service/Promatis.Net.MES.MDM.Service.csproj">
<Project Sdk="Microsoft.NET.Sdk">

    <PropertyGroup>
        <TargetFramework>net10.0</TargetFramework>
        <ImplicitUsings>enable</ImplicitUsings>
        <Nullable>enable</Nullable>
    </PropertyGroup>

    <ItemGroup>
        <ProjectReference Include="..\..\..\MesCore\Promatis.Net.MES.Service\Promatis.Net.MES.Service.csproj" />
        <ProjectReference Include="..\Promatis.Net.MES.MDM.Data\Promatis.Net.MES.MDM.Data.csproj" />
    </ItemGroup>

</Project>
</file>

<file path="Promatis.Net.MES.MDM.Service/Services/TechnologicalOperationParameterService.cs">
using Microsoft.EntityFrameworkCore;
using Promatis.Net.MES.MDM.Data;
using Promatis.Net.MES.MDM.Domain;
using Promatis.Net.MES.Service;

namespace Promatis.Net.MES.MDM.Service;

/// <summary>
/// BD8C0C'LC0KC' ACT@C$8D 4C'O D4?D 0C$;CT=C,,O C0@C,,2D07C=0CÄ8 C00D 0CÄ5D$@Cä2 C¢ >C05D 0Dd8D0<.D
/// </summary>
public class TechnologicalOperationParameterService(IDbContextFactory<MesMdmApplicationDbContext>
contextFactory)
: TechnologicalOperationParameterService<
    TechnologicalOperationParameter,

```

```

        TechnologicalOperation,Ø
        TechnologicalParameter,Ø
        MesMdmApplicationDbContext>(contextFactory)Ø
    {Ø
        // Aø;C AD 3CäBCä2 Cø 8D ?Cä;DÄ7Cä2C =C,,N C" 1C,,7C05D 0;Cä3C,,:CR 8 UI.Ø
    }
</file>

<file path="Promatis.Net.MES.MDM.Service/Services/TechnologicalOperationService.cs">
using Microsoft.EntityFrameworkCore;Ø
using Promatis.Net.MES.Data;Ø
using Promatis.Net.MES.MDM.Data;Ø
using Promatis.Net.MES.MDM.Domain;Ø
using Promatis.Net.MES.Service;Ø
using Promatis.Net.Service;Ø
Ø
namespace Promatis.Net.MES.MDM.Service;Ø
Ø
/// <summary>Ø
/// Aø>C05Dt=C 0 C0@C,,:C´0CD=C 0 D ;D46C 0 CD;D0 CC0@C 2C´5C08D0 BCTEC0>C´>C48Dt5D :C,,<C, >C05D 0Dd8D0<C,i
/// A,,!A0 A A´ A0 : At0Cø@D´2C 5D" BD >C"=Cä9 CD6CT=CT@C,,: C 1D BD 0CøBC0>C4> D 5D 2C,,AC OCD@C @CT0C´LC0KC
/// </summary>Ø
public class TechnologicalOperationService(IDbContextFactory<MesMdmApplicationDbContext>
contextFactory)Ø
    : TechnologicalOperationService<TechnologicalOperation, TechnologicalOperationUnit,
MesMdmApplicationDbContext>(contextFactory)Ø
{Ø
    /// <summary>Ø
    /// A0@C,,:C´0CD=C 0 DD0C @C,,:C ?Cä;C,,<Cä@DD8Ct<C ä
    /// Aø>C0AD$@D48D CCTB Dt8D BD´9 CD>Dt5D =C,,9 D,,0C ;Cä= C0> C$ACT< C0@C 2C,,;C < C# C, &W V–&VB0AC$>C"AD$2
    /// </summary>Ø
    public override Task<TechnologicalOperation> CreateChildTemplateAsync(TechnologicalOperation
parent)Ø
    {Ø
        if (parent == null) throw new ArgumentNullException(nameof(parent));Ø

        // At0D"8D$0 CD>CÄ5C00: CTAC´8 D >CD8D$5C´L D 0CÄ 0C$;D05D$AD0 ;C,,AD$>CÄÄ 3CT=CT@C,,@D45CÄ 8D :C´NDt5C
        if (parent.IsLeaf)Ø
        {Ø
            throw new InvalidOperationException(Ø
                $"A05C$>Ct<Cä6C0> D >Ct4C BDÄ 4CäGCT@C08C´ MC´5CÄ5C0B. Aä?CT@C FC,,0 '{parent.Name}' D02C´OCTB
            )Ø
        }

        // A,,=C,,FC,,0C´8Ct8D CCT< required-D 2Cä9D BC$0 D >C4;C AC0> D ?CTFC,,DC,,:C FC,,8 Cø>CÄ?C,,;D0BCä@C0>C4>
        var childOperation = new TechnologicalOperationØ
        {Ø
            Id = Guid.NewGuid(),Ø
            ParentId = parent.Id,Ø
            Parent = null, // B 0Ct@D´2C 5CÄ AD KC´:D2 2 Aä B2 4C´O Dt8D BCä9 UI-D 5D AC,,8D

            // A0> D4<Cä;Dt0C08Dä 2C0CD$@C, 3D CC0?D² ACä7CD0CT< Cø>C05Dt=D4N C BCä<C @C0CDä >C05D 0Dd8Dä ,, C
            IsLeaf = true,Ø

            Code = string.Empty,Ø
            Name = string.Empty,Ø
            Description = string.EmptyØ
        };Ø

        return Task.FromResult(childOperation);Ø
    }
}
</file>

<file path="Promatis.Net.MES.MDM.Service/Services/TechnologicalParameterCalcMethodService.cs">
using Microsoft.EntityFrameworkCore;Ø
using Promatis.Net.MES.Domain;Ø
using Promatis.Net.MES.MDM.Data;Ø
using Promatis.Net.MES.MDM.Domain;Ø
using Promatis.Net.MES.Service;Ø
Ø
namespace Promatis.Net.MES.MDM.Service;Ø
Ø
/// <summary>Ø
/// BD8C00C´LC0KC´ ACT@C$8D 4C´O D4?D 0C$;CT=C,,0 C,,=D BD CCøFC,,OCÄ8 C, <CTBCä4C <C, @C ADt5D$0 D$5C´5CÄ5D$@C
/// </summary>Ø
public class TechnologicalParameterCalcMethodService(IDbContextFactory<MesMdmApplicationDbContext>
contextFactory)Ø

```



```

        : TechnologicalParameterCalcMethodService<D
            TechnologicalParameterCalcMethod,D
            UnitBase,D
            TechnologicalOperation,D
            TechnologicalParameter,D
            MesMdmApplicationDbContext>(contextFactory)D
    {D
        // A0>C´=D´9 D ?CT:D$@ C 8Ct=CTA-C´>C48Cª8 CD;D0 =C AD$@Cä9Cª8 C 3D 5C40Dd8C, 4C =C0KDR 3CäBCä2.D
    }
</file>

<file path="Promatis.Net.MES.MDM.Service/Services/TechnologicalParameterService.cs">
using Microsoft.EntityFrameworkCore;D
using Promatis.Net.MES.Data;D
using Promatis.Net.MES.MDM.Data;D
using Promatis.Net.MES.MDM.Domain;D
using Promatis.Net.MES.Service;D
D
namespace Promatis.Net.MES.MDM.Service;D
D
/// <summary>D
/// BD8C00C´LC0KC´ ACT@C$8D 4C´O D 0C >D$K D > D ?D 0C$>Dt=C,,:Cä< D$5DT=Cä;Cä3C,,GCTACª8DR ?C @C <CTBD >C" 2
/// </summary>D
public class TechnologicalParameterService(IDbContextFactory<MesMdmApplicationDbContext>
contextFactory)D
    : TechnologicalParameterService<TechnologicalParameter,
MesMdmApplicationDbContext>(contextFactory)D
{D
    public override async Task<List<TechnologicalParameter>> GetAllAsync()D
    {D
        await using MesApplicationDbContext context = await ContextFactory.CreateDbContextAsync();D
    D
        return await context.Set<TechnologicalParameter>().D
            .Include(x => x.UnitOfMeasurement) // Ad0CD=C O Ct0C4@D47Cª0 D ?D 0C$>Dt=C,,:C CBä Ct<.D
            .ToListAsync();D
    }D
}
</file>

<file path="Promatis.Net.MES.MDM.Tests/Promatis.Net.MES.MDM.Tests.csproj">
<Project Sdk="Microsoft.NET.Sdk">D
D
    <PropertyGroup>D
        <TargetFramework>net10.0</TargetFramework>D
        <ImplicitUsings>enable</ImplicitUsings>D
        <Nullable>enable</Nullable>D
        <IsPackable>false</IsPackable>D
    </PropertyGroup>D
D
    <ItemGroup>D
        <PackageReference Include="coverlet.collector" Version="10.0.0">D
            <PrivateAssets>all</PrivateAssets>D
            <IncludeAssets>runtime; build; native; contentfiles; analyzers; buildtransitive</
IncludeAssets>D
        </PackageReference>D
        <PackageReference Include="FluentAssertions" Version="8.10.0" />D
        <PackageReference Include="FluentValidation" Version="12.1.1" />D
        <PackageReference Include="Microsoft.EntityFrameworkCore.InMemory" Version="10.0.8" />D
        <PackageReference Include="Microsoft.NET.Test.Sdk" Version="18.5.1" />D
        <PackageReference Include="Moq" Version="4.20.72" />D
        <PackageReference Include="xunit.runner.visualstudio" Version="3.1.5">D
            <PrivateAssets>all</PrivateAssets>D
            <IncludeAssets>runtime; build; native; contentfiles; analyzers; buildtransitive</
IncludeAssets>D
        </PackageReference>D
        <PackageReference Include="xunit.v3" Version="3.2.2" />D
    </ItemGroup>D
D
    <ItemGroup>D
        <ProjectReference Include="..
\Promatis.Net.MES.MDM.Service\Promatis.Net.MES.MDM.Service.csproj" />D
    </ItemGroup>D
D
    <ItemGroup>D
        <Using Include="Xunit" />D
    </ItemGroup>D

```

```
Ð
</Project>
</file>
```

```
<file path="Promatis.Net.MES.MDM.Tests/UnitCategoryValidationTests.cs">
```

```
using FluentAssertions;Ð
using FluentValidation.TestHelper;Ð
using Promatis.Net.MES.Domain.Interface;Ð
using Promatis.Net.MES.MDM.Domain;Ð
```

```
Ð
namespace Promatis.Net.MES.MDM.Tests;Ð
Ð
```

```
Ð
// B 5C ;C,,7C FC,,8 D CD"=CäAD$5C•
public class TestDept : DepartmentUnitBase { }Ð
public class TestProd : ProductionUnitBase { }Ð
public class TestStor : StorageUnitBase { }Ð
public class TestPos : PositionUnitBase { }Ð
public class TestTrans : TransportUnitBase { }Ð
Ð
```

```
// B 5C ;C,,7C FC,,8 C$0C`8CD0D$>D >C-
public class TestDeptValidator : DepartmentUnitBaseValidator<TestDept> { }Ð
public class TestProdValidator : ProductionUnitBaseValidator<TestProd> { }Ð
public class TestStorValidator : StorageUnitBaseValidator<TestStor> { }Ð
public class TestPosValidator : PositionUnitBaseValidator<TestPos> { }Ð
public class TestTransValidator : TransportUnitBaseValidator<TestTrans> { }Ð
Ð
```

```
public class UnitCategoryValidationTestsÐ
{Ð
```

```
    private readonly TestDeptValidator _deptValidator = new();Ð
    private readonly TestProdValidator _prodValidator = new();Ð
    private readonly TestStorValidator _storValidator = new();Ð
    private readonly TestPosValidator _posValidator = new();Ð
    private readonly TestTransValidator _transValidator = new();Ð
Ð
```

```
    [Theory]Ð
    [InlineData(UnitType.Workshop, true)] // Aä: CD;Dò CT?C @D$0CÄ5C0BC
    [InlineData(UnitType.Section, true)] // Aä: CD;Dò CT?C @D$0CÄ5C0BC
    [InlineData(UnitType.Crane, false)] // AäHC,,1C=0: A=0C = C05 CÄ>Cd5D" 1D`BDÂ CT?C @D$0CÄ5C0BCä<Ð
    public void DepartmentValidator_Should_Verify_Types(UnitType type, bool isValid)Ð
    {Ð
```

```
        var unit = new TestDept { Type = type, Name = "Dept" };Ð
        TestValidationResult<TestDept>? result = _deptValidator.TestValidate(unit);Ð
```

```
        if (isValid) result.ShouldNotHaveAnyValidationErrors();Ð
        else result.ShouldHaveValidationErrorFor(x => x).WithErrorMessage("B$8Cò =CR ACä>D$2CTBD BC$CCTB C=0D
```

```
    }Ð
```

```
    [Theory]Ð
    [InlineData(UnitType.MachineTool, true)] // Aä: CD;Dò &öGV7F-öí
    [InlineData(UnitType.Table, true)] // Aä: CD;Dò &öGV7F-öí
    [InlineData(UnitType.Storage, false)] // AäHC,,1C=0: B :C`0CB =CR <Cä6CTB C KD$! ProductionÐ
    public void ProductionValidator_Should_Verify_Types(UnitType type, bool isValid)Ð
    {Ð
```

```
        var unit = new TestProd { Type = type, Name = "Prod" };Ð
        TestValidationResult<TestProd>? result = _prodValidator.TestValidate(unit);Ð
```

```
        if (isValid) result.ShouldNotHaveAnyValidationErrors();Ð
        else result.ShouldHaveValidationErrorFor(x => x).WithErrorMessage("B$8Cò =CR ACä>D$2CTBD BC$CCTB C=0D
```

```
    }Ð
```

```
    [Theory]Ð
    [InlineData(UnitType.Cell, true)] // BôGCT9C=0 - DôBCâ C$ 4C`O B :C`0CD0Ð
    [InlineData(UnitType.Rack, true)] // B BCT;C`0Cb r MD$> Aä: CD;Dò !C=;C 4C
    [InlineData(UnitType.Vehicle, false)] // B$@C =D ?Cä@D" r =CR !C=;C 4Ð
    public void StorageValidator_Should_Verify_Types(UnitType type, bool isValid)Ð
    {Ð
```

```
        var unit = new TestStor { Type = type, Name = "Stor" };Ð
        TestValidationResult<TestStor>? result = _storValidator.TestValidate(unit);Ð
```

```
        if (isValid) result.ShouldNotHaveAnyValidationErrors();Ð
        else result.ShouldHaveValidationErrorFor(x => x).WithErrorMessage("B$8Cò =CR ACä>D$2CTBD BC$CCTB C=0D
```

```
    }Ð
```

```
    [Theory]Ð
    [InlineData(UnitType.Cell, true)] // BôGCT9C=0 CÄ>Cd5D" 1D`BDÂ Cä7C,,FC,,5C•
```

```

[InlineData(UnitType.Other, true)] // Other CÄ>Cd5D" 1D´BDÄ Cä7C,,FC,,5C•
[InlineData(UnitType.Workshop, false)] // Bd5DR =CR <Cä6CTB C KD$ L Aô>Ct8Dd8CT9D
public void PositionValidator_Should_Verify_Types(UnitType type, bool isValid)D
{D
    var unit = new TestPos { Type = type, Name = "Pos" };D
    TestValidationResult<TestPos>? result = _posValidator.TestValidate(unit);D
D
    if (isValid) result.ShouldNotHaveAnyValidationErrors();D
    else result.ShouldHaveValidationErrorFor(x => x).WithErrorMessage("B$8Cò =CR ACä>D$2CTBD BC$CCTB Cœ0D
}D
D
[Theory]D
[InlineData(UnitType.Vehicle, true)] // Aô>C4@D47Dt8C¢ r C¢ 4C´O B$@C =D ?Cä@D$0D
[InlineData(UnitType.Conveyor, true)] // Aœ>C02CT9CT@ – Aä: CD;Dò "D 0C0ACô>D BC
[InlineData(UnitType.MachineTool, false)] // B BC =Cä: – C05 B$@C =D ?Cä@D-
[InlineData(UnitType.Cell, false)] // BôGCT9Cœ0 – C05 B$@C =D ?Cä@D-
public void TransportValidator_Should_Verify_Types(UnitType type, bool isValid)D
{D
    var unit = new TestTrans { Type = type, Name = "Trans" };D
    TestValidationResult<TestTrans>? result = _transValidator.TestValidate(unit);D
D
    if (isValid) result.ShouldNotHaveAnyValidationErrors();D
    else result.ShouldHaveValidationErrorFor(x => x).WithErrorMessage("B$8Cò =CR ACä>D$2CTBD BC$CCTB Cœ0D
}D
D
D
[Fact]D
public void Kind_Should_Be_Automatically_Set_By_Base_Class()D
{D
    // Aô@Cä2CT@Dô5CÂÂ GD$> C00D,,8 C0>C$KCR 0C AD$@C :D$=D´5 Cœ;C AD K D 0CÄ8 Cô@CäAD$0C$;DôND" ¶-æBí
    // Aô5D 5CD0CT< Cä1Dô7C BCT;DÄ=D´9 Type C, æ ÖR 4C´O D >C ;Dä4CT=C,,O Cä3D 0C08Dt5C08C´ &W V–&VBí
D
    new TestDept { Type = UnitType.Other, Name = "Test" }.Kind.Should().Be(UnitKind.Department);D
    new TestProd { Type = UnitType.Other, Name = "Test" }.Kind.Should().Be(UnitKind.Production);D
    new TestStor { Type = UnitType.Other, Name = "Test" }.Kind.Should().Be(UnitKind.Storage);D
    new TestPos { Type = UnitType.Other, Name = "Test" }.Kind.Should().Be(UnitKind.Position);D
    new TestTrans { Type = UnitType.Other, Name = "Test" }.Kind.Should().Be(UnitKind.Transport);D
}D
}
</file>

<file path="Promatis.Net.MES.MDM.Tests/UnitTest1.cs">
namespace Promatis.Net.MES.MDM.TestsD
{D
    public class UnitTest1D
    {D
        [Fact]D
        public void Test1()D
        {D
D
        }D
    }D
}
</file>

<file path="Promatis.Net.MES.MDM.UI/_Imports.razor">
@using Microsoft.AspNetCore.Components.FormsD
@using Microsoft.AspNetCore.Components.RoutingD
@using Microsoft.AspNetCore.Components.WebD
@using MudBlazorD
@using Promatis.Net.UI.Components.WorkspaceD
@using Promatis.Net.UI.Components.EditDialogD
@using Promatis.Net.UI.Components.GridD
@using Promatis.Net.UI.Components.ToolbarD
@using Promatis.Net.UI.Components.Tree
</file>

<file path="Promatis.Net.MES.MDM.UI/Pages/TechnologicalOperations/Card/
TechnologicalOperationEditDialog.razor">
<EditDialog Title="@Title"D
    IsNew="@IsNew"D
    Model="@Model"D
    Validator="@Validator"D
    OnSaveAction="@OnSaveAction">D
<Tabs>D
    @* A$ A´ AD A ¢ D =Cä2C0KCR AC$>C"AD$2C BCTECô>C´>C48Dt5D :Cä9 Cä?CT@C FC,,8 *@D

```

```

<DialogTab Title="AäAC0>C$=Cä5" Icon="@Icons.Material.Filled.Info">D
  <Content>D
    <div class="d-flex flex-column gap-4 pt-2">D
      <MudTextField Label="A00C,,<CT=Cä2C =C,,5"D
        @bind-Value="Model.Name"D
        For="@(() => Model.Name)"D
        Margin="Margin.Dense"D
        Variant="Variant.Outlined"D
        Required="true" />D
      <MudTextField Label="B 8D BCT<C0KC' :Cä4"D
        @bind-Value="Model.Code"D
        For="@(() => Model.Code)"D
        Margin="Margin.Dense"D
        Variant="Variant.Outlined"D
        Required="true"D
        HelperText="B$>C'LC& C'0D$8C08Dd0 C" 2CT@DT=CT< D 5C48D BD 5, Dd8DD@D² 8 _
      <MudTextField Label="Aä?C,,AC =C,,5 / A00C,,<CTGC =C,,5"D
        @bind-Value="Model.Description"D
        For="@(() => Model.Description)"D
        Margin="Margin.Dense"D
        Variant="Variant.Outlined"D
        Lines="3" />D
      @* B4 B A$ AT A,, B "A "B4!Aä B$ B A,, A BÄ Aä Aä #At A ,, A,,!B$ ) B A )A,, "Aä AD AÄ
      <div class="d-flex align-center mt-2 pl-1">D
        <MudCheckBox T="bool"D
          Label="B02C'0CTBD 0 C&C05Dt=Cä9 Cä?CT@C FC,,5C' ,, C,,AD$>CÄ BCTECô@CäFCTA
          @bind-Value="Model.IsLeaf"D
          Disabled="@IsLeafModificationDisabled()"D
          Color="Color.Primary" />D
        </div>D
      </div>D
    </Content>D
  </DialogTab>D
  @* A$ A` AD A #0 C,,=C <C,,GCTAC&8C' ?Cä;C,,<Cä@DD=D´9 D 5C04CT@C,,=C2 ?C @C <CTBD >C" AC$0Ct8 *@D
  @if (Model.IsLeaf)D
  {D
    <DialogTab Title="B$5DT=Cä;Cä3C,,GCTAC&8CR ?C @C <CTBD K" Icon="@Icons.Material.Filled.Assignment"
    <Content>D
      <div class="d-flex flex-column gap-4 pt-2">D
        <MudText Typo="Typo.body2" Color="Color.Secondary" Class="italic">D
          B ?CTFC,,DC,,GC0KCR AC$0Ct8 Cä?CT@C FC,,8 D > D ?D 0C$>Dt=C,,:Cä< D$5DT=Cä;Cä3C,,GCTAC
        </MudText>D
        @* At4CTADÄ @C 7CÄ5D BC,,BD 0 C CCDCD'0D0 BC 1C'8Dd0/C4@C,,4 C00D 0CÄ5D$@Cä2 D 2D07C,
      </div>D
    </Content>D
  </DialogTab>D
  }D
</Tabs>D
</EditDialog>
</file>

```

```

<file path="Promatis.Net.MES.MDM.UI/Pages/TechnologicalOperations/Card/
TechnologicalOperationEditDialog.razor.cs">
using Microsoft.AspNetCore.Components;D
using Promatis.Net.MES.Domain;D
using Promatis.Net.MES.MDM.Domain;D
using Promatis.Net.UI.Components.Tree;D
D
namespace Promatis.Net.MES.MDM.UI.Pages.TechonologicalOperations.Card;D
D
public partial class TechnologicalOperationEditDialog : ComponentBaseD
{D
  [CascadingParameter] protected TreeActionContext<TechnologicalOperation> ActionContext { get;
set; } = null!;D
D
  [Parameter] public required string Title { get; set; }D
  [Parameter] public required bool IsNew { get; set; }D
  [Parameter] public required TechnologicalOperation Model { get; set; }D
  [Parameter] public required FluentValidation.Validator Validator { get; set; }D
  [Parameter] public required Func<Task> OnSaveAction { get; set; }D

```

```

Ð
/// <summary>Ð
/// A$KDt8D ;Dô5D"Â =D46CÔ> C´8 Ct0C ;Cä:C,,@Cä2C BDÂ 8Ct<CT=CT=C,,5 DD;C 3C -4ÆV b 2 UI CD;Dò 7C IC,,BD² 3
/// </summary>Ð
protected bool IsLeafModificationDisabled()Ð
{Ð
    // Aô@C, ACä7CD0C08C, =Cä2Cä3Cä MC´5CÄ5C0BC <CT=DôBDÂ AD$0D$CD <Cä6CÔ> D 2Cä1Cä4CÔ>Ð
    if (IsNew) return false;Ð
Ð
    // Aô@C, @CT4C :D$8D >C$0C08C, >Cô@C HC,,2C 5CÄ At#-C=MD, 1D >C=5D 0 CD0CÔ=D´E: CTAC´8 C$=D4BD 8 DÔBC
    // D46CR DC,,7C,,GCTAC=8 C´5Cd0D" 4CäGCT@C08CR >Cô5D 0Dd8C, r 1C´>C=8D CCT< C05D 5C=;Dä6C BCT;DÄ =C <CT
    if (ActionContext?.DataBroker?.InMemoryItems != null)Ð
    {Ð
        bool hasChildrenInCache = ActionContext.DataBroker.InMemoryItemsÐ
            .Any(x => x.ParentId == Model.Id);Ð
Ð
        // ATAC´8 CD5D$8 CTAD$L - C0@CT2D 0D$8D$L C" ;C,,AD" ,, -4ÆV b 0 G'VR´ 7C ?D 5D"5CÔ> CD>CÄ5CÔ=D´< CD
        return !TechnologicalOperationHierarchyEngine.CanChangeLeafStatus(hasChildrenInCache,
requestedIsLeaf: true);Ð
    }Ð
Ð
    return false;Ð
}Ð
}
</file>

<file path="Promatis.Net.MES.MDM.UI/Pages/TechnologicalOperations/TechnologicalOperationPage.razor">
@page "/mes/mdm/technological-operations"Ð
@using Promatis.Net.MES.MDM.DomainÐ
Ð
@inject TechnologicalOperationPageContext ContextÐ
Ð
<CascadingValue Value="@Context" IsFixed="true">Ð
    <WorkspacePage>Ð
Ð
        <TopContent>Ð
            @* B$#A´ A : B 0D ?Cä;C 3C 5D$ADò AC$5D ED2Â =C BC,,2CÔ> CÄ0D HC ;C,,@D45D" :Cä<C =CDK C" :Cä=D$5C
            <UiToolbar TEntity="TechnologicalOperation"Ð
                ActionContext="@Context"Ð
                OnCreateTriggered="@Context.OnCreateRootOrSiblingActionAsync"Ð
                OnCreateChildTriggered="@Context.OnCreateChildActionAsync"Ð
                OnEditTriggered="@Context.OnUpdateActionAsync"Ð
                OnDeleteTriggered="@Context.OnDeleteActionAsync" />Ð
            </TopContent>Ð
Ð
            <BodyContent>Ð
                @* AD B A$ : Aô5D 5CD0CT< D 5C´5C=BCä@ D$5C=AD$0 D :D 0D 8C$KCÂ >D$>C @C 6CT=C,,5CÄ AD$0D$CD 0 C
                <TreePage TEntity="TechnologicalOperation"Ð
                    NodeTextSelector="@ (x => $"[{x.Code}] {(x.IsLeaf ? "0=Ý9" : "0=ÚÁ")} {x.Name} )" />Ð
                </BodyContent>Ð
Ð
            </WorkspacePage>Ð
        </CascadingValue>
    </file>

<file path="Promatis.Net.MES.MDM.UI/Pages/TechnologicalOperations/
TechnologicalOperationPageContext.cs">
using FluentValidation;Ð
Ð
using MudBlazor;Ð
using Promatis.Net.Domain.Interface;Ð
using Promatis.Net.MES.MDM.Domain;Ð
using Promatis.Net.MES.MDM.UI.Pages.TechonologicalOperations.Card;Ð
using Promatis.Net.MES.Service;Ð
Ð
using Promatis.Net.UI.Components.Tree;Ð
Ð
namespace Promatis.Net.MES.MDM.UI.Pages.TechonologicalOperations;Ð
Ð
public class TechnologicalOperationPageContext : TreeActionContext<TechnologicalOperation>Ð
{Ð
    private readonly ITechnologicalOperationService<TechnologicalOperation,
TechnologicalOperationUnit> _operationService;Ð
    private readonly IDialogService _dialogService;Ð
    private readonly IValidator<TechnologicalOperation> _globalValidator;Ð
    private readonly IEntityCloner _entityCloner;Ð

```

```

Ð
/// <summary>Ð
/// AD AÔ AÂ Bt B A,, B B 'AT" AD B "B4 AÔ B "A, AÔ AÔ A, Aä A A' AÔ Bò Aä B4 A' .Ð
/// AÔ0D$8C$=Cä >CÔ@C HC,,2C 5D" 4Cä<CT=CÔKC' 4C$8Cd>C£¢ 5D ;C, 2D'1D 0CÔ=C O Cä?CT@C FC,,0 Dô2C'0CTBD O A'
/// Cä=Cä?C¢0 "AD>C 0C$8D$L CÔ>CDCct5C²" 0C$BCä<C BC,,GCTAC¤8 C40D =CTB CÔ0 D$CC'1C @CR T'Â 7C IC,,IC O B #
/// </summary>Ð
public override bool IsCreateChildEnabled =>Ð
    SelectedData != null && !SelectedData.IsLeaf;Ð
Ð
public TechnologicalOperationPageContext(Ð
    ITechnologicalOperationService<TechnologicalOperation, TechnologicalOperationUnit>
operationService,Ð
    IDialogService dialogService,Ð
    IValidator<TechnologicalOperation> globalValidator,Ð
    IEntityCloner entityCloner) : base()Ð
{Ð
    _operationService = operationService ?? throw new
ArgumentNullException(nameof(operationService));Ð
    _dialogService = dialogService ?? throw new ArgumentNullException(nameof(dialogService));Ð
    _globalValidator = globalValidator ?? throw new
ArgumentNullException(nameof(globalValidator));Ð
    _entityCloner = entityCloner ?? throw new ArgumentNullException(nameof(entityCloner));Ð
Ð
    PageTitle = "B ?D 0C$>Dt=C,,: D$5DT=Cä;Cä3C,,GCTAC¤8DR >Cô5D 0Dd8C'½
}Ð
Ð
/// <summary>Ð
/// B 5C ;C,,7C FC,,0 Cä1Dô7C BCT;DÄ=Cä3Cä <CTBCä4C ?D >C4@CT2C At#-C¤MD,,0 C @Cä:CT@C 4C =CÔKDR1
/// A$KctKC$0CTBD O C 2D$>CÄ0D$8Dt5D :C, ?D 8 C,,=C,,FC,,0C'8Ct0Dd8C, :Cä<Cô>CÔ5CÔBC G&VU vR 1CT7 C ;Cä:C,,
/// </summary>Ð
public override async Task InitializeInMemoryTreeAsync()Ð
{Ð
    List<TechnologicalOperation> data = await _operationService.GetAllAsync();Ð
    DataBroker.InMemoryItems = data;Ð
}Ð
Ð
// =====Ð
// B A A,, A &A,,/ CRUD-Aä AT A &A,, AD Bò Aä A AB "B4 A B (Aô B$"AT AÔ !Aä!AT A Aô AD#At A •
// =====Ð
Ð
/// <summary>Ð
/// A$KctKC$0CTBD O D$CC'1C @Cä< CÔ@C, =C 6C BC,,8 C¤=Cä?C¤8 "B >Ct4C BDÂ" ,, Cä2D'9 C¤>D 5CÔL C,,;C, ACäACT
/// </summary>Ð
public async Task OnCreateRootOrSiblingActionAsync()Ð
{Ð
    TechnologicalOperation targetUnit;Ð
Ð
    // B &AT A A,, A ¢ CD5D 5C$5 CÔ8Dt5C4> CÔ5 C$KC @C =Cä r ACä7CD0CT< Dt8D BD'9 A¤ B AÔ, (A4@D4?CÔ8
    if (SelectedData == null)Ð
    {Ð
        targetUnit = new TechnologicalOperationÐ
        {Ð
            Id = Guid.NewGuid(),Ð
            IsLeaf = false, // Aô> D4<Cä;Dt0CÔ8Dä :Cä@CT=DÂ r MD$> CÔ0Cô:C Ô3D CCô?C
            ParentId = null,Ð
            Parent = null,Ð
            Code = string.Empty,Ð
            Name = string.Empty,Ð
            Description = string.EmptyÐ
        };Ð
Ð
        await OpenEditDialogAsync(targetUnit, "B >Ct4C =C,,5 C¤>D =CT2Cä9 C4@D4?CôK Cä?CT@C FC,,9", isNew:
return;Ð
    }Ð
Ð
    // B &AT A A,, A ¢ CD5D 5C$5 C$KC @C = DÔ;CT<CT=D" r ACä7CD0CT< D >D 5CD0 B /AD AÂ ,,=C BCä< Cd5 D
    if (SelectedData.Parent != null)Ð
    {Ð
        // Aô@CäAC,,< DD0C @C,,:D2 1DÔ:CT=CD0 D >Ct4C BDÂ GC,,AD$KC' HC 1C'>CÔ 4CäGCT@CÔ5C4> DÔ;CT<CT=D$0 CD
        targetUnit = await _operationService.CreateChildTemplateAsync(SelectedData.Parent);Ð
        targetUnit.Parent = SelectedData.Parent; // A$>D AD$0CÔ0C$;C,,2C 5CÄ AD KC':D2 =C @Cä4C,,BCT;Dò 4C
    }Ð
    elseÐ
    {Ð
        // ATAC'8 C$KC @C =CÔKC' MC'5CÄ5CÔB D 0CÄ :Cä@CT=DÂÂ 7CÔ0Dt8D" 5C4> D >D 5CB r MD$> D$>Cd5 C¤>D 5
        targetUnit = new TechnologicalOperationÐ

```

```

        {
            IsLeaf = false,
            ParentId = null,
            Parent = null,
            Code = string.Empty,
            Name = string.Empty,
            Description = string.Empty
        };
    }

    targetUnit.Id = Guid.NewGuid();

    string dialogTitle = SelectedData.Parent != null
        ? $"B >Ct4C BDÂ >C05D 0Dd8DÂ =C CD >C$=CR wµ6VÆV7FVDF F Å &VçBäæ ÖW0r-
        : "B >Ct4C =C,,5 Cα>D =CT2Cä9 Cä?CT@C FC,,8";

    await OpenEditDialogAsync(targetUnit, dialogTitle, isNew: true);
}

/// <summary>
/// A$KctKC$0CTBD 0 D$CC'1C @Cä< C0@C, =C 6C BC,,8 Cα=Cä?Cα8 "AD>C 0C$8D$L C0>CDCct5C²"í
/// </summary>
public async Task OnCreateChildActionAsync(TechnologicalOperation parent)
{
    if (parent == null) return;

    // At0C0@C HC,,2C 5CÂ ?D 5CD7C ?Cä;C05C0=D'9 D,,0C ;Cä= D2 DC 1D 8Cα8 D 5D 2C,,AC0>C4> D ;Cä0 C MCα5C04C
    TechnologicalOperation childUnit = await _operationService.CreateChildTemplateAsync(parent);
    childUnit.Id = Guid.NewGuid();
    childUnit.Parent = parent; // A0@C,,=D44C,,BCT;DÄ=Cä AC$0CtKC$0CT< C" At# CD;D0 :Cä=D$5CαAD$0 CD8C ;Cä

    await OpenEditDialogAsync(childUnit, $"AD>C 0C$8D$L C0>CDCct5C² 2 '{parent.Name}'", isNew: true);
}

/// <summary>
/// A$KctKC$0CTBD 0 D$CC'1C @Cä< C0@C, =C 6C BC,,8 Cα=Cä?Cα8 "A,,7CÄ5C08D$L".D
/// </summary>
public async Task OnUpdateActionAsync(TechnologicalOperation node)
{
    if (node == null) return;

    // A$KctKC$0CT< Dd5C0BD 0C'8ct>C$0C0=D'9 CD2C,,6Cä: C4;D41Cä:Cä3Cä :C'>C08D >C$0C08D0 OCD@C ?C'0D$DCä
    var targetClone = _entityCloner.CloneEntity(node);

    // A$>D AD$0C00C$;C,,2C 5CÂ =C 2C,,3C FC,,>C0=D'5 D4:C 7C BCT;C, =C 6C,,2Cä5 CD5D 5C$> C" >C05D 0D$8C$=C
    targetClone.ParentId = node.ParentId;
    targetClone.Parent = node.Parent;

    await OpenEditDialogAsync(targetClone, editTitle: $"B 5CD0CαBC,,@Cä2C =C,,5: {node.Name}", isNew: false);
}

/// <summary>
/// A$KctKC$0CTBD 0 D$CC'1C @Cä< C0@C, =C 6C BC,,8 Cα=Cä?Cα8 "B44C ;C,,BDÂ"í
/// </summary>
public async Task OnDeleteActionAsync(TechnologicalOperation node)
{
    if (node == null) return;

    // ATAC'8 D2 ?C ?Cα8 CTAD$L C$;Cä6CT=C0KCR MC'5CÄ5C0BD² 2 Aä B20:D0HCRÄ 2D'2Cä4C,,< C0@CT4D4?D 5Cd4CT=
    bool hasChildren = DataBroker.InMemoryItems?.Any(x => x.ParentId == node.Id) ?? false;
    string confirmationMessage = hasChildren
        ? $"A$=C,,<C =C,,5! A4@D4?C00 '{node.Name}' D >CD5D 6C,,B C$;Cä6CT=C0KCR >C05D 0Dd8C,â #CD0C'8D$L CT
        : $"A$K D42CT@CT=D²Ä GD$> DT>D$8D$5 D44C ;C,,BDÄ BCTEC0>C'>C48Dt5D :D4N Cä?CT@C FC,,N '{node.Name}'

    var parameters = new DialogParameters { ["ContentText"] = confirmationMessage };
    var options = new DialogOptions { CloseOnEscapeKey = true };

    var dialog = await _dialogService.ShowAsync<MudMessageBox>("B44C ;CT=C,,5 Cä?CT@C FC,,8", parameters, options);
    var result = await dialog.Result;

    if (result is not null && !result.Canceled)
    {
        await _operationService.DeleteAsync(node.Id);
    }
}
}

```

```

// =====
// A$+At A" #AÔ A$ B !A BÃ Aä Aä A,, A´ A4 BD B B½
// =====
private async Task OpenEditDialogAsync(TechnologicalOperation model, string editTitle, bool
isNew)
{
    var parameters = new DialogParameters
    {
        ["Title"] = editTitle,
        ["IsNew"] = isNew,
        ["Model"] = model,
        ["Validator"] = _globalValidator,
        ["OnSaveAction"] = async () =>
        {
            if (isNew)
            {
                // A A,"A,"'AT!A AR B B A$ AT A,, : AÔCÄ5D BC$> Ct0CÔCC´OCT< CÔ0C$8C40Dd8Cä=CÔCDâ AD
                // BÔBCâ 1C´>C8D CCTB D 5C=CD AC,,2CÔKC´ >C ECä4 EF Core C, ?Cä;CÔ>D BDÄN C´8C=2C,,4C,,@D45
                // A 0Ct0 CD0CÔ=D´E D 2Dô6CTB C$5D$:C, ?Cä ?C´>D :Cä<D2 ?Cä;Dâ &VçD-BÃ :CäBCä@Cä5 Ct0CÔ
                model.Parent = null;
            }

            await _operationService.AddAsync(model);
        }
        else
        {
            // AD;Dò >C =Cä2C´5CÔ8Dò =C 2C,,3C FC,,>CÔ=D4N D AD´;C= C D$>Cd5 Ct0CÔCC´OCT<, DtBCä1D² 8Ct1
            model.Parent = null;
        }

        await _operationService.UpdateAsync(model);
    }
};

var options = new DialogOptions { CloseButton = true, MaxWidth = MaxWidth.Small, FullWidth
= true };
await _dialogService.ShowAsync<TechnologicalOperationEditDialog>(editTitle, parameters,
options);
}
</file>

<file path="Promatis.Net.MES.MDM.UI/Pages/TechnologicalParameter/Card/
TechnologicalParameterDialog.razor">
<EditDialog Title="@Title"
    IsNew="@IsNew"
    Model="@Model"
    Validator="@Validator"
    OnSaveAction="@OnSaveAction">
    <Tabs>
        @* A AÄ Aä AT B" A A A 1 *@
        <DialogTab Title="AäACÔ>C$=Cä5" Icon="@Icons.Material.Filled.Info">
            <Content>
                <TechnologicalParameterGeneralTab Model="@Model" />
            </Content>
        </DialogTab>
        @* A AÄ Aä AT B" A A A 2 *@
        <DialogTab Title="A>CÔDC,,3D4@C FC,,O DCA" Icon="@Icons.Material.Filled.Settings">
            <Content>
                <TechnologicalParameterTechnicalTab Model="@Model" />
            </Content>
        </DialogTab>
    </Tabs>
</EditDialog>
</file>

<file path="Promatis.Net.MES.MDM.UI/Pages/TechnologicalParameter/Card/
TechnologicalParameterDialog.razor.cs">
using FluentValidation;
using Microsoft.AspNetCore.Components;
using Promatis.Net.MES.Domain.Interface;
namespace Promatis.Net.MES.MDM.UI.Pages.TechnologicalParameter.Card;
public partial class TechnologicalParameterDialog : ComponentBase
{

```



```

[Parameter] public required string Title { get; set; }
[Parameter] public required bool IsNew { get; set; }
[Parameter] public required Domain.TechnologicalParameter Model { get; set; }
[Parameter] public required IValidator Validator { get; set; }
[Parameter] public required Func<Task> OnSaveAction { get; set; }
}
</file>

<file path="Promatis.Net.MES.MDM.UI/Pages/TechnologicalParameter/Card/
TechnologicalParameterGeneralTab.razor">
@using Promatis.Net.MES.MDM.Domain
@
<div class="d-flex flex-column gap-4 pt-2">
    <MudTextField Label="A00C,,<CT=Câ2C =C,,5"
        @bind-Value="Model.Name"
        For="@(() => Model.Name)"
        Margin="Margin.Dense"
        Variant="Variant.Outlined"
        Required="true" />
    @
    <MudTextField Label="B 8D BCT<C0KC' :Câ4"
        @bind-Value="Model.Code"
        For="@(() => Model.Code)"
        Margin="Margin.Dense"
        Variant="Variant.Outlined"
        Required="true"
        HelperText="B$>C'LC C'0D$8C08Dd0 C" 2CT@DT=CT< D 5C48D BD 5, Dd8DD@D² 8 _" />
    @
    <MudTextField Label="Aä?C,,AC =C,,5 / A0@C,,<CTGC =C,,5"
        @bind-Value="Model.Description"
        For="@(() => Model.Description)"
        Margin="Margin.Dense"
        Variant="Variant.Outlined"
        Lines="3" />
</div>
@
@code {
    [Parameter] public required TechnologicalParameter Model { get; set; }
}
</file>

<file path="Promatis.Net.MES.MDM.UI/Pages/TechnologicalParameter/Card/
TechnologicalParameterTechnicalTab.razor">
@using Promatis.Net.Domain.Interface
@using Promatis.Net.MES.Domain
@using Promatis.Net.MES.Domain.Interface
@
<div class="d-flex flex-column gap-4 pt-2">
    @
    @* A$+A0 AD Bä)A,, B A,,!A B B0 AT B Aä Aä AT . A,, AÄ B A0 B0 B$ A0 AÄ A A0+BR
    <MudSelect T="Guid?"
        Label="AT4C,,=C,,FC 8Ct<CT@CT=C,,0"
        @bind-Value="Model.UnitOfMeasurementId"
        For="@(() => Model.UnitOfMeasurementId)"
        Margin="Margin.Dense"
        Variant="Variant.Outlined"
        Required="true">
        @foreach (UnitOfMeasurement uom in _availableUoms)
        {
            <MudSelectItem Value="@((Guid?)uom.Id)">@($"[{uom.Code}] {uom.Name}")</MudSelectItem>
        }
    </MudSelect>
    @
    <MudSelect T="string"
        Label="B$8C0 4C =C0KDR ?C @C <CTBD 0"
        @bind-Value="Model.DataType"
        For="@(() => Model.DataType)"
        Margin="Margin.Dense"
        Variant="Variant.Outlined"
        Required="true">
        @foreach (string type in AllowedDataTypes)
        {
            <MudSelectItem Value="@type">@type</MudSelectItem>
        }
    </MudSelect>
    @

```

```

@* AÄ Aä AT!B$ AT AÔ+A' B' Aä AD Bò Aä A,, "Aä Aä Aä A4 A A,, 'AT A,,/ B AT BT# *@
<MudSelect T="CalculationMethod"
    Label="B 0Ct@CTHCT=CÔKCR <CTBCä4D² 0C4@CT3C FC,,8 DCA"
    MultiSelection="true"
    SelectedValues="_selectedMethods"
    SelectedValuesChanged="OnAllowedMethodsChanged"
    Margin="Margin.Dense"
    Variant="Variant.Outlined"
    HelperText="A$KC 5D 8D$5 CöCC² <CTBCä4Cä2, CD>D BD4?CÔKDR 4C'0 Cö@Cä8Ct2Cä4D BC$0">
@foreach (CalculationMethod method in Enum.GetValues(typeof(CalculationMethod)))
{
    @if (method != CalculationMethod.None)
    {
        <MudSelectItem Value="@method">@method.GetDescription()</MudSelectItem>
    }
}
</MudSelect>
}
</div>
</file>

<file path="Promatis.Net.MES.MDM.UI/Pages/TechnologicalParameter/Card/
TechnologicalParameterTechnicalTab.razor.cs">
using Microsoft.AspNetCore.Components;
using Promatis.Net.MES.Domain;
using Promatis.Net.MES.Domain.Interface;
using Promatis.Net.MES.Service;
namespace Promatis.Net.MES.MDM.UI.Pages.TechonologicalParameter.Card;
public partial class TechnologicalParameterTechnicalTab : ComponentBase
{
    [Inject] protected IUnitOfMeasurementService UomService { get; set; } = null!;
    [Parameter] public required Domain.TechonologicalParameter Model { get; set; }

    protected readonly string[] AllowedDataTypes = ["Numeric", "String", "Boolean", "DateTime"];
    protected List<UnitOfMeasurement> _availableUoms = [];

    // A,,!Aô A A' Aô : Aô5D 5C$5CD5C0> C00 C²>C0BD 0C²B IReadOnlyCollection D >C4;C AC0> D$@CT1Cä2C =C,,0CÄ ×
    protected IReadOnlyCollection<CalculationMethod> _selectedMethods =
Array.Empty<CalculationMethod>();

    protected override async Task OnInitializedAsync()
    {
        await base.OnInitializedAsync();
        _availableUoms = await UomService.GetAllAsync();

        // B 0D ?C :Cä2D'2C 5CÄ GC,,AC'> Cö>C 8D$>C$>C' <C AC²8 C,,7 A C" 4CäAD$CCô=D'9 CÄ0D AC,,2 DD;C 3Cä2D
        _selectedMethods = Enum.GetValues(typeof(CalculationMethod))
            .Cast<CalculationMethod>()
            .Where(method => method != CalculationMethod.None &&
Model.AllowedMethods.HasFlag(method))
            .ToList()
            .AsReadOnly(); // Aô@C,,2CT4CT=C,,5 C$ •&V DöæÇ"6öÆV7F-öí

    }

    /// <summary>
    /// B @C 1C BD'2C 5D" ?D 8 C,,7CÄ5C05C08C, 3C ;CäGCT:.
    /// A,,!Aô A A' Aô : A @C4CCÄ5C0B Cö@C,,2CT4CT= C$ BC,,?D2 •&V DöæÇ"6öÆV7F-öâ 4C'0 D4AD$@C =CT=C,,0 CäHC,,1C
    /// </summary>
    protected void OnAllowedMethodsChanged(IReadOnlyCollection<CalculationMethod> updatedValues)
    {
        _selectedMethods = updatedValues;

        // B EC'>CÔKC$0CT< CÄ0D AC,,2 Dt5C²1Cä:D >C" >C @C BC0> C" ?Cä1C,,BCä2Cä5 Dt8D ;Cä -çB 4C'0 B #A
        CalculationMethod resultMask = CalculationMethod.None;
        foreach (var method in updatedValues)
        {
            resultMask |= method;
        }

        Model.AllowedMethods = resultMask;
    }
}
</file>

```

```

<file path="Promatis.Net.MES.MDM.UI/Pages/TechnologicalParameter/TechnologicalParameterPage.razor">
@page "/mes/mdm/technological-parameters"
@
@using Promatis.Net.MES.MDM.Domain
@using Promatis.Net.UI
@* A$=CT4D OCT< C00D, 0C$BCä<C BC,,GCTAC=8 Ct0D 5C48D BD 8D >C$0C0=D´9 C" D´ :Cä=D$5C=AD" ´?Cä;C0>D$K C$;C AD$
@inject TechnologicalParameterPageContext Context
@
@* A0@Cä:C,,4D´2C 5CÄ :Cä=D$5C=AD" :C AC=0CD>CÄ 2C08CrÂ GD$>C K DT>C´AD" v÷&·7 6U vR =C BC,,2C0> Dt8D$0C² =C
<CascadingValue Value="@Context" IsFixed="true">
    <WorkspacePage>
        <TopContent>
            <UiToolbar TEntity="TechnologicalParameter"
                ActionContext="@Context"
                OnCreateTriggered="@Context.OnCreateActionAsync"
                OnEditTriggered="@Context.OnUpdateActionAsync"
                OnDeleteTriggered="@Context.OnDeleteActionAsync" />
        </TopContent>
        <BodyContent>
            @* B$ A A,,&A ¢ Cä;Cä=C=8 D 5C04CT@D0BD O C" :C =Cä=C,,GC0>CÄ BCT3CR 6öçVÖç46öçFVçB ?C´0D$DCä@CÄK
            <GridPage TEntity="TechnologicalParameter">
                <ColumnsContent>
                    <PropertyColumn T="TechnologicalParameter" TProperty="string"
                        Property="x => x.Code"
                        Title="B 8D BCT<C0KC´ :Cä4"D
                        HeaderStyle="width: 150px;"
                        Class="font-monospace" />
                    <PropertyColumn T="TechnologicalParameter" TProperty="string"
                        Property="x => x.Name"
                        Title="A00C,,<CT=Cä2C =C,,5 C00D 0CÄ5D$@C -
                        HeaderStyle="width: 300px;" />
                    @* A,,!A0 A A´ A0 : B GC,,BD´2C 5CÄ :D 0D$:C,,9 C CC=2CT=C0KC´ :Cä4 C,,7 D 2D07C =C0>C4> A0!
                    <PropertyColumn T="TechnologicalParameter" TProperty="string"
                        Property="x => x.UnitOfMeasurement != null ?
x.UnitOfMeasurement.Code : string.Empty"
                        Title="AT4. C,,7CÄâ-
                        HeaderStyle="width: 100px;" />
                    <PropertyColumn T="TechnologicalParameter" TProperty="string"
                        Property="x => x.DataType"
                        Title="B$8C0 4C =C0KDR-
                        HeaderStyle="width: 120px;">
                        <CellTemplate>
                            <MudChip T="string" Color="@context.Item.DataType.GetActionColor()"
Size="Size.Small" Variant="Variant.Text">
                                @context.Item.DataType
                            </MudChip>
                        </CellTemplate>
                    </PropertyColumn>
                    <PropertyColumn T="TechnologicalParameter" TProperty="string"
                        Property="x => x.Description"
                        Title="Aä?C,,AC =C,,5 / A0@C,,<CTGC =C,,5" />
                </ColumnsContent>
            </GridPage>
        </BodyContent>
    </WorkspacePage>
</CascadingValue>
</file>

<file path="Promatis.Net.MES.MDM.UI/Pages/TechnologicalParameter/
TechnologicalParameterPageContext.cs">
using MudBlazor;
using Promatis.Net.Domain.Interface;
using Promatis.Net.MES.MDM.Service;
using Promatis.Net.MES.MDM.UI.Pages.TechonologicalParameter.Card;
using Promatis.Net.UI.Components.Grid;
@
namespace Promatis.Net.MES.MDM.UI.Pages.TechonologicalParameter;

```

```

    }
    /// <summary>
    /// A<C0BCT:D B D4?D 0C$;CT=C,,0 C0@C,,:C'0CD=Cä9 D BD 0C08Dd5C' AC0@C 2CäGC08C=D D$5DT=Cä;Cä3C,,GCTAC=8DR ?C @
    /// A0>C'=CäAD$LDä >D 2Cä1Cä6CD5C0 >D" 2C,,7D40C'LC0>C4> boilerplate-C=D0 C, CC0@C 2C'0CTB D CC4CC > C'>C48C
    /// </summary>
    public class TechnologicalParameterPageContext : GridActionContext<Domain.TechnologicalParameter>
    {
        private readonly TechnologicalParameterService _parameterService;
        private readonly IDialogService _dialogService;
        private readonly IEntityCloner _entityCloner;

        public TechnologicalParameterPageContext(
            TechnologicalParameterService parameterService,
            IDialogService dialogService,
            IEntityCloner entityCloner)
        {
            _parameterService = parameterService ?? throw new
ArgumentNullException(nameof(parameterService));
            _dialogService = dialogService ?? throw new ArgumentNullException(nameof(dialogService));
            _entityCloner = entityCloner ?? throw new ArgumentNullException(nameof(entityCloner));

            PageTitle = "B$5DT=Cä;Cä3C,,GCTAC=8CR ?C @C <CTBD K";

            IsCreateVisible = true;
            IsEditVisible = true;
            IsDeleteVisible = true;

            DataBroker.ConfigureInMemoryMode();
        }

        public async Task InitializeInMemoryDataAsync()
        {
            try
            {
                List<Domain.TechnologicalParameter> allParameters = await
_parameterService.GetAllAsync();
                DataBroker.InMemoryItems = allParameters;
            }
            catch (Exception ex)
            {
                Console.WriteLine($"[ERROR] AäHC,,1C=D C,,=C,,FC,,0C'8Ct0Dd8C, At#-D ?D 0C$>Dt=C,,:C ?C @C <CTBD >C#
            }
        }

        public Task OnCreateActionAsync()
        {
            var newParameter = new Domain.TechnologicalParameter();

            // A0@D0<Cä9 C$Kct>C" >C >C ICT=C0>C4> CÄ5D$>CD0 D04D 0 D CC=0Ct0C08CT< C=D0:D 5D$=Cä9 Blazor-DD>D
            return OpenEditDialogAsync<TechnologicalParameterDialog>(
                newParameter,
                "B >Ct4C =C,,5 D$5DT=Cä;Cä3C,,GCTAC=D>C4> C00D 0CÄ5D$@C "í
                isNew: true,
                saveDelegate: async () =>
                {
                    // BT8D CD 3C,,GCTAC=8 C,,7Cä;C,,@D45CÄ =C 2C,,3C FC,,>C0=D'9 C4@C D C05D 5CB Tb 6÷&]
                    newParameter.UnitOfMeasurement = null;
                    await _parameterService.AddAsync(newParameter);
                }
            );
        }

        public Task OnUpdateActionAsync()
        {
            if (SelectedItem == null) return Task.CompletedTask;

            Domain.TechnologicalParameter targetClone = _entityCloner.CloneEntity(SelectedItem);
            targetClone.UnitOfMeasurementId = SelectedItem.UnitOfMeasurementId;
            targetClone.UnitOfMeasurement = SelectedItem.UnitOfMeasurement;

            // A0@D0<Cä9 C$Kct>C" >C >C ICT=C0>C4> CÄ5D$>CD0 D04D 0 CD;D0 @CT4C :D$8D >C$0C08D0 AD$@Cä:C, ?C @C <
            return OpenEditDialogAsync<TechnologicalParameterDialog>(
                targetClone,
                $"B 5CD0C=BC,,@Cä2C =C,,5: {targetClone.Name}",
                isNew: false,

```

```

        saveDelegate: async () =>{
            {
                targetClone.UnitOfMeasurement = null;
                await _parameterService.UpdateAsync(targetClone);
            }
        });
    }

    public async Task OnDeleteActionAsync()
    {
        if (SelectedItem == null) return;

        bool? confirm = await _dialogService.ShowDialogAsync(
            "B44C ;CT=C,,5 Ct0C08D 8",
            $"A$K CD5C'AD$2C,,BCT;DÄ=Cä ECäBC,,BCR CCD0C'8D$L D$5DT=Cä;Cä3C,,GCTACp8C' ?C @C <CTBD wµ6VÆV7FVD-F",
            yesText: "B44C ;C,,BDÄ"Ä 6 æ6VÄFW†C$ $ D$<CT=C ""½

        if (confirm == true)
        {
            await _parameterService.DeleteAsync(SelectedItem.Id);
        }
    }
}
</file>

```

```

<file path="Promatis.Net.MES.MDM.UI/Promatis.Net.MES.MDM.UI.csproj">
<Project Sdk="Microsoft.NET.Sdk.Razor">
    <PropertyGroup>
        <TargetFramework>net10.0</TargetFramework>
        <Nullable>enable</Nullable>
        <ImplicitUsings>enable</ImplicitUsings>
    </PropertyGroup>

    <ItemGroup>
        <SupportedPlatform Include="browser" />
    </ItemGroup>

    <ItemGroup>
        <PackageReference Include="Microsoft.AspNetCore.Components.Web" Version="10.0.8" />
        <PackageReference Include="MudBlazor" Version="9.4.0" />
    </ItemGroup>

    <ItemGroup>
        <ProjectReference Include="..\..\..\Core\Configuration.Web\Promatis.Net.Configuration.Web.csproj" />
        <ProjectReference Include="..\..\..\MesCore\Promatis.Net.MES.UI\Promatis.Net.MES.UI.csproj" />
        <ProjectReference Include="..\Promatis.Net.MES.MDM.Service\Promatis.Net.MES.MDM.Service.csproj" />
    </ItemGroup>
</Project>
</file>

```

```

<file path="Promatis.Net.MES.MDM.UI/UiModule.cs">
using MudBlazor;
using Promatis.Net.UI;
namespace Promatis.Net.MES.MDM.UI;

public class UiModule : IUiModule
{
    public string Name => GetType().Assembly.GetName().Name ?? "Unknown.Module";

    public IEnumerable<(string Title, string Href, string Icon, string? Group)> GetMenuItems()
    {
        return new List<(string Title, string Href, string Icon, string? Group)>
        {
            // B ?D 0C$>Dt=C,,: D$5DT=Cä;Cä3C,,GCTACp8DR >C05D 0Dd8C•
            ("B$5DT=Cä;Cä3C,,GCTACp8CR >C05D 0Dd8C,"Ä "öÖW2öÖF0÷FV6†æöÆöv-6 ÄÖ÷ W& F-öç2"Ä
            Icons.Material.Filled.SettingsInputComponent, "B ?D 0C$>Dt=C,,:C,"Í

            // Aö;CäACp8C' ACô@C 2CäGC08C$ BCTECô>C'>C48Dt5D :C,,E C00D 0CÄ5D$@Cä2D
            ("B$5DT=Cä;Cä3C,,GCTACp8CR ?C @C <CTBD K", "/mes/mdm/technological-parameters",
            Icons.Material.Filled.SettingsInputComponent, "B ?D 0C$>Dt=C,,:C,"•
        }
    }
}

```

```
        };  
    }  
</file>  
  
<file path="Promatis.Net.MES.MDM.UI/wwwroot/exampleJsInterop.js">  
    // This is a JavaScript module that is loaded on demand. It can export any number of  
    // functions, and may import other JavaScript modules if required.  
      
    export function showPrompt(message) {  
        return prompt(message, 'Type anything here');  
    }  
</file>  
  
</files>
```